

1. Administrative & Study Identification

Full Study Title: Real-World Treatment Study of Koselugo (Selumetinib) **Short Title/Acronym:** Koselugo Real-World Study (Korea) **Protocol Number:** D1346R00009 **NCT Number:** NCT06360406 **Sponsor Name:** AstraZeneca **Investigational Product:** Koselugo (Selumetinib) **Phase of Development:** Phase N/A (Observational Post-Approval Study) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To characterize the safety and effectiveness of Koselugo (Selumetinib) in routine clinical practice settings in South Korea, specifically assessing the known safety profile (including physeal dysplasia) and identifying any previously unsuspected adverse reactions. **Secondary Objectives:** To evaluate the effectiveness of Koselugo via descriptive analysis of physician qualitative assessments regarding the overall disease status of Neurofibromatosis type 1 (NF-1) and clinically significant plexiform neurofibromas (PNs).

2.2 Background & Rationale As part of a post-approval commitment requested by the Korean health authority, this study is designed to generate real-world evidence for patients treated with Koselugo (Selumetinib), an oral selective inhibitor of MAPK kinase (MEK) 1 and 2. It aims to provide critical information on the Korean patient population managed under conditions of routine daily medical practice.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Observational (Real-World Treatment Study) **Control Type:** Single-arm **Blinding:** Open-label (Routine Care) **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target N\$:** 200 participants **Number of Sites:** 9 locations across South Korea **Estimated Study Start:** 15-Jun-2024 **Estimated Primary Completion:** 30-Sep-2031

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Patients treated with Koselugo (Selumetinib) under the approved label in South Korea. 2.

Provision of signed and dated written informed consent by the patient or a legally acceptable representative.

4.2 Exclusion Criteria 1. Patients who do not meet the inclusion criteria or decline to provide informed consent. (*Note: As a real-world observational study, no additional strict protocol-defined exclusions apply beyond prescribing label contraindications.*)

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Prescribed according to the approved product label and treating physician's clinical discretion. **Route of Administration:** Oral **Duration of Treatment:** Routine clinical practice duration, with structured effectiveness and safety data evaluated over a 1-year timeframe per outcome measure.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications for comorbidities as prescribed by the treating physician in routine care. **Prohibited:** Any therapies strictly contraindicated in the locally approved Koselugo product label in South Korea.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening/Baseline	Day 0 Informed Consent, Demographics, Medical History, Confirmation of Koselugo Prescription
2	Follow-Up Visits	Routine Practice Standard clinical assessments, Ongoing Safety monitoring (AEs/SAEs), Physician qualitative assessment of NF-1/PNs
3	End of Assessment	1 Year Final safety tracking (AEs, physeal dysplasia) and effectiveness qualitative assessments

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 1. Adverse events (AEs) rate: Includes AEs, serious AEs (SAEs), adverse drug reactions (ADRs), serious ADRs (SADRs), and unexpected AEs/ADRs. 2. Physeal dysplasia occurrence rate. **Measurement Method:** Clinical observation and safety reporting over a 1-year timeframe.

7.2 Secondary & Exploratory Endpoints Definition: Descriptive analysis with physician qualitative assessments regarding the overall disease status of NF-1 and the status of clinically significant PNs. **Measurement Method:** Status categorized as Improving, Progression, or Stable over a 1-year timeframe.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via routine clinical practice and post-marketing surveillance tracking.

Serious Adverse Events (SAEs): Reported within standard local regulatory timelines to health authorities and the sponsor.

Data Monitoring Committee (DMC): Overseen by the sponsor's standard pharmacovigilance and real-world evidence team parameters.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All enrolled patients providing informed consent who receive at least one dose of Koselugo.

Sample Size Justification: $N = 200$, powered to detect safety signals and establish real-world effectiveness within the targeted South Korean patient population, aligned with post-approval commitments.

Handling of Missing Data: As a real-world evidence study, data will be analyzed using descriptive statistics reflecting routine clinical practice documentation without aggressive imputation methodologies.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Pharmacoeepidemiology Practices (GPP) and local Korean health authority regulations for post-approval safety studies.

Monitoring Plan: Post-marketing surveillance verification aligning with the local routine care framework. **Audit Trail:** eCRF locking and privacy-compliant health data abstraction through standard secure databases.

11. Study Identifiers & Governance

Primary ID: D1346R00009 **Registry Number:** NCT06360406 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** Real-World Treatment Study of Koselugo (Selumetinib) **Official Regulatory Title:** Real-World Treatment Study of Koselugo (Selumetinib)

12. Clinical Framework (Neurofibromatosis 1 Specific)

Primary Condition: Neurofibromatosis 1 (NF-1) **Diagnosis**

Threshold: Diagnosis of NF-1 with symptomatic, inoperable Plexiform Neurofibromas (PNs). **Medical Methodology:** MEK 1 and 2 inhibitor therapy per approved indication. **Optimization Target:** Disease stabilization (Stable, Improving) and monitoring of physeal dysplasia.

13. Study Procedures & Methodology

Management Pathway: Standard routine daily medical practice for NF-1 in South Korea. **Education Protocol (HCP):** Familiarization with Koselugo local approved label and safety reporting standards. **Education Protocol (Patient):** Informed consent administration and routine care disease education. **Quality Audit Mechanism:** Chart reviews and data extraction from real-world health records.

14. Observation & Follow-Up Schedule

Total Duration: Individual evaluation at 1-year mark; Study continues overall until Sept 2031. **Onsite Visit Cadence:** Dictated by standard of care and physician discretion. **Remote Monitoring Frequency:** Routine pharmacovigilance timelines. **Checklist Review Interval:** Routine intervals corresponding to clinical follow-ups.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				